Tan Dai Ngo

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Technical Skills

- **Programming & Data Engineering:** Python (Numpy, Pandas), SQL, Java, MATLAB, Cassandra, Neo4j, Kafka
- **Machine Learning & Data Science:** supervised and unsupervised learning, multi-output regression, gradient boosted decision trees (XGBoost, CatBoost), scikit-learn, feature engineering, cross-validation, model evaluation
- **Deployment:** Docker, FastAPI, AWS (EC2), CI/CD (Jenkins), model serialization, log monitoring (Splunk)

Education

Master of Engineering in Applied Data Science — **September 2025 - December 2026 (Expected)**
University of Victoria

- **GPA:** 9.0 / 9.0
- **Relevant Coursework:** Optimization for Machine Learning, Data Analysis and Pattern Recognition, Systems for Massive Datasets, Information Theory, Algorithms and Data Models

Master of Science in Computer Science — University of Chicago

Bachelor of Science in Economics — University of Washington

Experience

Software Developer | T-Mobile (via BeaconFire Inc.) — *Bellevue, WA, USA* | **June 2022 - April 2024**

- Collaborated with a team of 5 to develop enterprise microservices for the Roaming Business System (Spring Boot, Java Stream API), integrating with Jenkins CI/CD for deployments and Splunk for log monitoring.
- Designed and optimized Cassandra database schemas for partner and workflow management microservices, handling 100k+ records while maintaining query execution time under 3 seconds.
- Automated weekly reprocessing of 50+ roaming service tests with Kafka pipelines, cutting manual effort by 80%.

Projects

End-to-End Machine Learning System for Multi-Output Fuel Blending — Shell.ai Hackathon 2025

- Deployed a production ready machine learning system using FastAPI, Docker, and AWS EC2 to serve real time multi-output regression predictions for 10 chemical blend properties.
- Benchmarked XGBoost vs. CatBoost using 5-fold cross-validation, selecting CatBoost after achieving lower error (MAPE 0.64 vs. 1.29) and productionizing the best performing model.
- Engineered robust feature and inference pipelines using weighted property aggregation, entropy based mixture metrics, and serialized preprocessing to ensure consistent production inference.

Stock Sentiment Analysis (Team of 4) — Algorithms and Data Models Project

- Built a sentiment analysis pipeline over 15k+ tweets across 10 tech stocks, enabling large-scale market sentiment analysis, by integrating X API data, SQLite/Neo4j storage, and FinTwitBERT modeling.
- Identified measurable sentiment-return relationships, including statistically significant predictive effects for NVIDIA, by implementing engagement-weighted sentiment aggregation and Granger causality testing.
- Analyzed equity volatility dynamics across major tech stocks, revealing persistent volatility clustering, by applying GARCH(1,1) modeling and automated cross-stock analysis workflows.

Handwritten Digit Recognition System — Optimization for Machine Learning Project

- Developed a multiclass SVM and Softmax pipeline using a custom built ML-BFGS optimizer, achieving 97.65% accuracy with faster convergence than standard SGD.
- Leveraged HOG and PCA to reduce 784 pixel dimensions to 50 latent components, resulting in a 10.2% accuracy lift and 40% faster training latency.
- Optimized regularization parameters (μ, C) and evaluated performance via 10*10 confusion matrices to ensure robust generalization on 10,000 test samples.